

Explanation Stability and Post-hoc Trust Governance for Class-Incremental Network Traffic Classification

OFRA prediction with an offline ETG evidence ledger

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Abstract

Class-incremental network-traffic classification must learn newly introduced classes without an oracle task identifier while retaining earlier knowledge. We evaluate prediction, explanation stability, and post-hoc governance as linked but separately bounded layers. OFRA combines a shared tabular encoder, family-specific low-rank binary heads, bounded replay, and centroid-based routing. Five-seed prediction results across five independently processed datasets show dataset-dependent trade-offs. A same-data Balanced Replay50 comparison finds Malaya accuracy of 54.68% versus 40.14% and signed forgetting of 3.55 versus 43.80 percentage points. On ReplayIDS D2, primary OFRA improves accuracy and Macro-F1 but lowers attack recall from 99.78% to 85.73%; the comparison does not isolate architecture from loss, sampling, or memory differences. A separate five-seed Malaya analysis compares Expected Gradients, Feature Ablation, and Gradient x Input on a registered routed class margin. Their mean silent-drift rates are 73.98%, 3.60%, and 14.34%, with all-method agreement of 50.0% for admission, 54.0% for ETG state, and 35.0% for drift. ETG is consequently a method-conditioned offline ledger, not an independent safety certificate. Task-0-only normalisation and vocabulary fitting are distinguished from offline dataset construction. Cross-environment ReplayIDS score reconstruction fails a fixed numerical-fidelity gate, limiting portability claims. The evidence establishes measurable retention, detection, and explanation trade-offs rather than universal superiority, autonomous unknown discovery, or an online feedback loop.

Keywords: class-incremental learning; network traffic classification; explainable AI; explanation drift; SHAP; trust governance; intrusion detection.

1. Introduction

1.1 Background and motivation

Network traffic changes as applications, services, attacker behaviour, and collection environments evolve. A fixed classifier therefore becomes incomplete, but repeatedly retraining one shared model can overwrite knowledge of earlier classes. Class-incremental learning addresses this setting by presenting new classes sequentially and evaluating all classes observed so far. For a security-facing system, predictive retention is only one requirement. Analysts may also rely on explanations to understand why a class score was produced, so a stable accuracy value does not by itself guarantee stable decision evidence.

This study distinguishes prediction, explanation, and governance. OFRA produces the class decision. SHAP explains a registered class margin at stored checkpoints; the primary Malaya analysis uses fixed true-class probes. ETG consumes performance and explanation evidence after inference and writes a governance ledger. An explanation-quality failure does not cancel the underlying classification alert. ETG has not been evaluated as a controller of the learner.

1.2 Problem statement

Current class-incremental NIDS evaluation commonly prioritises accuracy and forgetting, whereas explanation stability and operational response are less consistently measured. A valid audit must explain the same score used by the registered inference path; an approximate wrapper or a separate surrogate model cannot establish governance of the deployed decision. The present study therefore uses a source-bound analysis of OFRA's registered joint-cap score. The remaining problem is to determine whether the current predictor retains useful performance across increments, whether its explanation ranking changes under performance-preserved transitions, and how uncertain or unstable explanations should be recorded without overstating deployment maturity.

1.3 Research questions and objectives

RQ1: How does oracle-free class-incremental learning affect predictive performance, class-balanced detection, and forgetting across heterogeneous network-traffic datasets?

RQ2: To what extent do prediction explanations remain stable across successive class-incremental updates when class-level predictive performance remains stable?

RQ3: How can explanation instability be converted into auditable governance decisions without altering or suppressing the underlying security prediction?

RO1 is to evaluate accuracy, Macro-F1, balanced accuracy, attack recall, false-positive rate, and forgetting under independently preprocessed five-seed class-incremental protocols. RO2 is to measure explanation stability on fixed probes across adjacent checkpoints using source-bound attribution, Top-k feature overlap, and class-performance constraints. RO3 is to implement and evaluate an offline evidence ledger that records certification, refusal, escalation, and re-certification decisions from prediction and attribution evidence.

1.4 Contributions and attribution of methods

The study-specific contribution is the design and evaluation of a class-incremental predictor together with an audit of its routed class margins and an explanation-governance ledger that leaves prediction active. The experiments separate retention, minority-class coverage, attribution stability, and governance state, rather than treating a single accuracy or explanation score as sufficient evidence. The five-seed sensitivity measurements and negative results define where this design is useful and where it remains inadequate.

The constituent learning and explanation methods are not new algorithms introduced here. Low-rank adaptation, FT-Transformer, TabM, DP-Means, SHAP, Feature Ablation, and Gradient x Input are existing methods. Family-head integration, the registered score combination, fixed-probe audit protocol, and ledger rules are study-specific design choices. A combination of established components does not by itself demonstrate algorithmic novelty or a causal architectural advantage. Those stronger claims require controlled component comparisons and operational validation. FT-Transformer follows Gorishniy et al. (2021); DP-Means originates from Kulis and Jordan (2012).

2. Related Work

2.1 Continual learning and parameter-efficient adaptation

Catastrophic forgetting motivates regularisation, rehearsal, distillation, exemplar, and parameter-isolation methods in continual learning (French, 1999; Kirkpatrick et al., 2017; Rebuffi et al., 2017). LoRA freezes a base transformation and represents a trainable update with low-rank matrices (Hu et al., 2022). For tabular representation learning, TabM uses parameter-efficient ensembling to produce multiple predictions within one model (Gorishniy et al., 2025), while CatBoost is a gradient-boosted tree method designed to reduce prediction shift in boosting (Prokhorenkova et al., 2018). The present family-specific low-rank head and TabM mean-embedding adapter are project integration choices; neither is presented as a new derivation of the cited methods.

2.2 Explanation drift and IDS governance

Recent work has moved beyond one-time post-hoc explanation. E2D2 detects explanation-based drift in unsupervised NIDS, but it targets distributional drift monitoring rather than legitimate class-incremental parameter updates (Nugraha and Bauschert, 2026). Maseno et al. (2026) use SHAP instability as an auxiliary signal for adversarial evasion, whereas the present study excludes adversarial perturbation and compares consecutive learning checkpoints. Goncalves and Alomari (2026) evaluate analyst-facing SHAP and LIME explanations in cloud intrusion detection, but do not define a class-incremental explanation ledger. Adabara et al. (2025) combine SHAP, probability calibration, and uncertainty in a cognitive IDS, yet do not test non-suppressing governance actions after explanation drift. These studies establish closely related motivation, but none closes the specific gap between legitimate class expansion, the exact routed decision score, and a governance record that leaves the predictive alert active.

Table 1. Positioning against the four recent studies identified during review.

Study	Drift or trust trigger	IDS setting	Gap relative to this work
E2D2 (2026)	Unsupervised explanation-based drift	Unsupervised NIDS	Not legitimate class-incremental model updates; no ETG action ledger
Maseno et al. (2026)	Adversarial SHAP instability	Recurrent multiclass IDS	Adversarial trigger, not benign learning increments; no non-suppressing governance
Goncalves and Alomari (2026)	Human interpretation of SHAP/LIME	AWS GuardDuty	User study rather than longitudinal routed-score stability
Adabara et al. (2025)	Calibration, uncertainty, and SHAP	CICIDS2017	No class-incremental sequence or explanation-state transition ledger

2.3 Research gap

The research target is a source-bound protocol that explains the registered routed score, measures stability under legitimate class-incremental updates, and records non-suppressing governance states. The cited studies motivate this specific combination but do not establish its uniqueness across all literature. We evaluate the combination and its limitations without claiming that explanation stability proves prediction correctness or security effectiveness.

3. Methodology

3.1 Scope and dataset separation

Each dataset is preprocessed, trained, and evaluated independently; records from different datasets are never pooled. The strict five-seed table uses seeds 1, 2, 3, 4, and 42. NSL-KDD, UNSW-NB15, CIC-IDS-2017, and CSE-CIC-IDS2018 use FT-Transformer 256x4; MalayaNetwork_GT uses FT-Transformer 512x12. These are dataset-specific audits, not a capacity-matched pooled comparison. The CSE-CIC-IDS2018 8+10-epoch campaign completed as DICC Job 402073 and passed the protected SHA-256 manifest check.

Dataset-specific preprocessing removes identifiers, normalises labels, converts features to finite numerical arrays, fixes feature and class order, and constructs class-by-split shards. NSL-KDD retains its official train/test files; its protocol, service, and flag vocabularies are fitted on Task-0 training rows only and frozen, producing 116 features. UNSW-NB15 likewise retains its official 175,341-row training and 82,332-row testing files; its protocol, service, and state vocabularies are fitted on Task-0 training rows only and frozen, producing 186 features. Later unseen categorical values use the registered all-zero block for that field. Malaya uses 31 derived-flow CSVs at frozen revision 384a5927..., removes IP addresses, ports, and timestamps, and holds out one capture per class, producing 77 numerical features. The capture-level Malaya split avoids leakage from correlated rows within one capture.

The current formal prediction protocols use Task-0-only numerical statistics and categorical vocabularies. Numerical standardisation reads only Task-0 training shards, freezes before pretraining, and rejects later updates. NSL-KDD and UNSW-NB15 vocabularies likewise use Task-0 rows only; later unseen values map to all-zero categorical blocks. The strict widths are 116 and 186 features, and the affected five-seed experiments were rerun. Earlier 122/194-feature builders exposed six and eight future-only one-hot columns, affecting about 0.9% of later-task rows. That schema exposure is historical rather than a property of the current prediction table. CIC-IDS-2017, CSE-CIC-IDS2018, and Malaya have no learned categorical vocabulary. This fitting contract does not imply that every separate dataset-construction policy is prospective; the ReplayIDS D2 exception is specified below.

3.2 OFRA architecture and registered decision arms

For Malaya, the shared FT-Transformer maps each numerical row to a 512-dimensional representation using 12 layers, 16 attention heads, head dimension 32, and dropout 0.1. The other formal datasets use the separately registered FT256x4 configuration. After Task-0 training the shared encoder is frozen. Task 0 uses eight epochs and later tasks ten epochs. Each seen class has a rank-8, alpha-16 positive-versus-negative low-rank head. Replay retains at most 50 exemplars per class from a candidate pool of at most 5,000.

The DP-Means router stores up to 32 centroids per class. The capped router fits centroids from at most 3,000 selected training samples per class; the cap limits router fitting cost and majority-class influence, not test-set size. The registered arms are head-only $p(c,x)$, router-only $z(c,x)$, and joint $p(c,x)+0.5z(c,x)$, each with capped

and full router variants. Prediction is the global argmax across all seen classes. The joint weight 0.5, cap 3,000, distance quantile 0.3, and centroid limit 32 are study-defined registered settings rather than universal constants.

3.3 Training, metrics, and statistical boundary

The reported FT-Transformer configuration uses focal loss (gamma 2.0, alpha 0.75), learning rate 0.001, training batch size 384, evaluation batch size 512, and a negative-to-positive sampling ratio of four. At every checkpoint, evaluation covers all classes seen so far. Accuracy measures all correct predictions; Macro-F1 averages class F1 equally; balanced accuracy averages class recall; forgetting is the mean reduction from each earlier task's best historical accuracy to its later value. Current formal prediction values use seeds 1, 2, 3, 4, and 42. Model capacity and schedule are disclosed per dataset, and conclusions are descriptive unless a paired interval is explicitly reported.

3.4 Additive classifier-comparison protocol

TabM is added as a minimally invasive OFRA encoder comparison (Gorishniy et al., 2025). It produces 16 member embeddings of width 256 and averages them to the single representation required by the existing family-head interface. The family heads, exemplar budget, DP-Means routers, joint weight 0.5, cap 3,000, task order, and metric definitions remain unchanged. This is a mean-embedding adapter rather than a member-wise TabM-OFRA redesign.

CatBoost is evaluated separately as a cumulative multiclass diagnostic (Prokhorenkova et al., 2018). At each checkpoint it retrains on all training rows from classes seen so far. It does not use OFRA family heads, bounded replay, DP-Means routing, or ETG, and its native feature attribution is not the registered routed-margin SHAP target. The comparison therefore estimates classification headroom in the fixed feature representation rather than a matched continual-learning effect.

3.5 SHAP explanation protocol

The analyzer uses `shap.GradientExplainer`, an expected-gradients approximation within the SHAP framework (Lundberg and Lee, 2017). It reconstructs the registered joint-cap class margin (the explained class's joint score minus the largest competing seen-class joint score) and attributes that margin to input features. Positive attributions support the explained margin and negative attributions oppose it. Top-15 features are ranked by absolute attribution magnitude. Fixed probes and backgrounds are source-bound and reused across adjacent checkpoints. Source alignment is distinct from numerical equivalence in a new runtime; the separate ReplayIDS portability check is reported in Section 4.10.

For adjacent checkpoints, top-feature overlap is the size of the intersection divided by the size of the union of the two feature sets. The registered silent explanation-drift rule requires the class recall to fall by no more than 5 percentage points while top-15 Jaccard falls below 0.70. Both thresholds are project governance settings rather than externally validated standards. The event unit is a class-by-adjacent-checkpoint transition, not a packet, flow, sample, or real-world incident.

3.6 ETG governance scope

ETG is an offline explanation-governance ledger. It can certify an admission, refuse insufficient explanation evidence, escalate a drifted transition, and require strict recertification. A failed explanation action withholds the explanation alert or routes the evidence for review; it does not cancel OFRA's underlying classification alert. The completed five-seed analysis consumes stored Malaya OFRA checkpoint evidence for seeds 1, 2, 3, 4, and 42. ETG has therefore been connected to the actual routed-score evidence path, but it has not been evaluated as a closed loop that changes future routing, training, or replay.

3.7 Attribution-method robustness protocol

The source-bound robustness campaign uses Malaya checkpoints for seeds 1, 2, 3, 4, and 42, fixed true-class probes, the joint-cap-3,000 class margin, the top-15 rule, the registered random-control admission threshold, and the same ETG state machine. Its three primary rankings are Expected Gradients, single-feature ablation to the frozen checkpoint mean, and Gradient x Input. Pairwise top-15 Jaccard, admission agreement, ETG-state agreement, and transition-level silent-drift agreement are reported per seed and summarized with descriptive Student-t intervals. Integrated Gradients remain a failed single-seed numerical diagnostic and are excluded from the primary five-seed comparison. This design measures method dependence rather than assuming that any explainer is uniquely correct.

3.8 No-look-ahead audit protocol

The audit traces every data-dependent preprocessing state to its fitting split and task availability. It checks numerical accumulator reads and freeze points, categorical-vocabulary construction, official-test access, and row-level impact of future-only categories. The audit distinguishes future-category schema exposure from future-label training and from test leakage, then reports exact affected columns and rows rather than a binary pass/fail label.

3.9 Paired checkpoint-selection protocol

A ReplayIDS-aligned CIC-IDS-2017 D2 experiment compares last-epoch selection with two training-only checkpoint rules under the same fixed split, seeds {1, 2, 3, 4, 42}, FT256x4 encoder, replay-50 budget, training trajectories, and official joint-cap-3,000 inference. The unconstrained candidate selects the earliest maximum binary Macro-F1 on the training-only calibration holdout. The guarded candidate uses the same candidate checkpoints but admits a non-final checkpoint only when calibration Macro-F1 improves by at least 0.01, attack-recall loss is at most 0.01, and benign-FPR increase is at most 0.01; these are project-defined operating guards, not published universal thresholds. Heartbleed has one calibration row and therefore uses the registered last-epoch fallback. The five seeds are paired repetitions on one data split, not five independent datasets.

3.10 Same-data replay comparison and offline construction

Balanced Replay50 updates a shared FT-Transformer and multiclass head, pairing each later-task new row with one draw from old-class memory and retaining 50 herded examples per observed class. It is compared with same-seed OFRA joint-cap-3,000 results on Malaya and a separate ReplayIDS-aligned CIC-IDS-2017 D2 derivative. Splits, class supports, Task-0-only normalisation, and task order match within each dataset. Equal exemplar counts and epochs do not match total memory or compute: OFRA also stores centroids, replay changes row exposure, and the shared multiclass comparator differs from OFRA's binary focal loss and negative sampling. This is a system-level comparison, not an architecture-only ablation.

D2 caps normal-class training rows at the largest attack-class fitting count, 124,780. It contains 268,697 fitting rows, 68,313 training-only calibration rows, and 227,723 fixed test rows with 78 features. The cap uses counts from all fitting attack classes, including later increments. Thus D2 is an offline benchmark-construction policy, not a fully prospective no-look-ahead sampling policy. Official-test labels do not fit the cap, normaliser, or checkpoint selector. D2 results must not replace the distinct CIC-IDS-2017 results in Table 3.

Seeds 1, 2, 3, 4, and 42 vary training randomness on a fixed split and class order. Seed 42 was used to select the replay comparator in an earlier screen, so the all-five summary is descriptive and seeds 1-4 are also reported separately. ReplayIDS uses last-epoch OFRA checkpoints as the primary comparator; guarded checkpoint selection remains secondary. Exact two-sided paired tests with five nonzero pairs cannot attain $p < 0.05$ (minimum 0.0625). Neither seed replication nor retrospective re-analysis constitutes five independent datasets.

4. Results

4.1 RQ2: Explanation stability

Across five seeds, Expected Gradients reports 61 silent-drift events among 82 eligible class-by-adjacent-checkpoint transitions. The mean seed-level rate is 73.98% with a descriptive 95% Student-t interval of 58.38% to 89.57%. The earlier seed-1 value of 12/17 (70.59%) is retained only as a historical component of this aggregate. The evidence unit is a registered class transition, not a packet, flow, sample, or real-world incident.

Under the registered Expected-Gradients rule, stable class recall does not guarantee stable top-feature membership. However, the five-seed comparison also shows that this conclusion is not invariant to the attribution method. Feature Ablation reports 3/82 events and Gradient x Input reports 12/82. The result does not establish that the model became unsafe, that one explainer is wrong, or that any percentage of network flows drifted. The class-recall and Jaccard thresholds remain study-defined operating settings rather than universal standards.

4.2 RQ2: Attribution-method robustness

Across the five stochastic seeds, the three primary methods agree on admission for 50.0% of checkpoint-class rows on average, on ETG state for 54.0%, and on the silent-drift conclusion for 35.0% of adjacent transitions. Expected Gradients, Feature Ablation, and Gradient x Input have mean seed-level silent-drift rates of 73.98%, 3.60%, and 14.34%, respectively. Mean pairwise top-15 Jaccard is 0.516, 0.357, and 0.381 for the three method pairs. The result confirms material attribution

dependence: ETG outcomes are method-conditioned, and the registered Expected-Gradients result is not treated as uniquely correct.

Table 2. Five-seed attribution-method agreement on the source-bound Malaya routed score.

Method pair	Top-15 Jaccard	Admission agreement	ETG-state agreement	Drift agreement
Expected vs ablation	0.516	68.7%	69.3%	40.0%
Expected vs Grad x Input	0.357	64.0%	69.3%	43.0%
Ablation vs Grad x Input	0.381	67.3%	63.3%	87.0%

Integrated Gradients are excluded from Table 2 because their completeness check failed (mean row-level error 11.58; maximum 373.72). They are retained as a failed numerical diagnostic. The three valid rankings quantify sensitivity but do not determine which explainer is causally correct.

4.3 RQ1: Predictive performance and retention

Table 3. Current strict five-seed prediction evidence for the two main decision arms.

Dataset / model	Arm	Accuracy	Macro-F1	Balanced acc.	Forgetting
NSL-KDD FT256x4	Head only	56.29	31.08	36.48	8.14 pp
NSL-KDD FT256x4	Joint cap	71.56	41.46	42.25	1.60 pp
UNSW-NB15 FT256x4	Head only	68.78	21.16	23.52	2.68 pp
UNSW-NB15 FT256x4	Joint cap	61.95	23.50	28.45	8.97 pp
CIC-IDS-2017 FT256x4	Head only	64.59	21.25	25.91	9.04 pp
CIC-IDS-2017 FT256x4	Joint cap	72.57	36.95	63.13	9.63 pp
Malaya FT512x12	Head only	56.03	11.77	15.05	3.16 pp
Malaya FT512x12	Joint cap	54.68	21.15	23.03	3.55 pp
CSE-CIC-IDS2018 FT256x4	Head only	79.21	19.06	21.15	2.00 pp
CSE-CIC-IDS2018 FT256x4	Joint cap	50.16	33.55	52.15	19.07 pp

The routing effect is dataset-dependent. On NSL-KDD, joint cap 3,000 improves accuracy, Macro-F1, balanced accuracy, and forgetting. On UNSW-NB15, it improves Macro-F1 and balanced accuracy but lowers overall accuracy and increases forgetting. On CIC-IDS-2017, it improves accuracy and class-balanced metrics but does not reduce forgetting. On Malaya, the paired five-seed difference is -1.34 accuracy points, +9.37 Macro-F1 points, +7.97 balanced-accuracy points, and +0.39 forgetting points; pooled per-class results show non-zero recall for all ten classes under joint cap, but several minority recalls remain low. The table therefore supports dataset-specific trade-offs, not universal superiority.

The strict CSE-CIC-IDS2018 FT256x4 campaign completed five registered seeds with eight Task-0 epochs and ten epochs per later task. Head-only achieved 79.21% accuracy, 19.06% Macro-F1, 21.15% balanced accuracy, 2.00 points of forgetting, 31.70% attack recall, and 6.57% benign false-positive rate. Joint cap-3000 achieved 50.16%, 33.55%, 52.15%, 19.07 points, 91.82%, and 51.85%, respectively. Joint uncapped achieved 54.39%, 33.72%, 51.35%, 15.39 points, 84.38%, and 46.48%. Thus routing recovers minority/attack coverage but produces an operationally severe accuracy, retention, and false-positive cost on this dataset. The earlier 1+1-epoch closure remains historical and is not mixed with this result.

4.4 RQ1: Additive classifier comparison

Table 4. Malaya classifier comparisons. Values are mean +/- sample SD across five seeds; protocols remain separate.

Model and protocol	Accuracy	Macro-F1	Balanced acc.	Forgetting
TabM-OFRA joint cap, 5 seeds	57.99 +/- 3.93	20.46 +/- 1.19	21.60 +/- 1.70	4.39 +/- 2.56 pp
FT512-OFRA joint cap, 5 seeds	54.68 +/- 2.71	21.15 +/- 3.37	23.03 +/- 3.42	3.55 +/- 0.77 pp
CatBoost cumulative, 5 seeds	66.50 +/- 0.44	34.76 +/- 0.69	39.10 +/- 0.73	15.34 +/- 0.73 pp
TabM cumulative, 5 seeds	65.01 +/- 0.96	32.66 +/- 1.31	34.41 +/- 1.51	9.43 +/- 1.80 pp

Across the same five seeds, TabM-OFRA joint-cap accuracy is 57.99% versus 54.68% for FT512-OFRA, a +3.31-point difference. Macro-F1 is 20.46% versus 21.15% (-0.69 points), balanced accuracy is 21.60% versus 23.03% (-1.43 points), and forgetting is 4.39 versus 3.55 points (+0.84 points). The TabM change therefore improves aggregate accuracy but not the class-balanced or retention metrics. Because BitTorrent is the majority class, this result is not evidence of uniform minority-class improvement.

4.5 RQ3: ETG governance outcomes

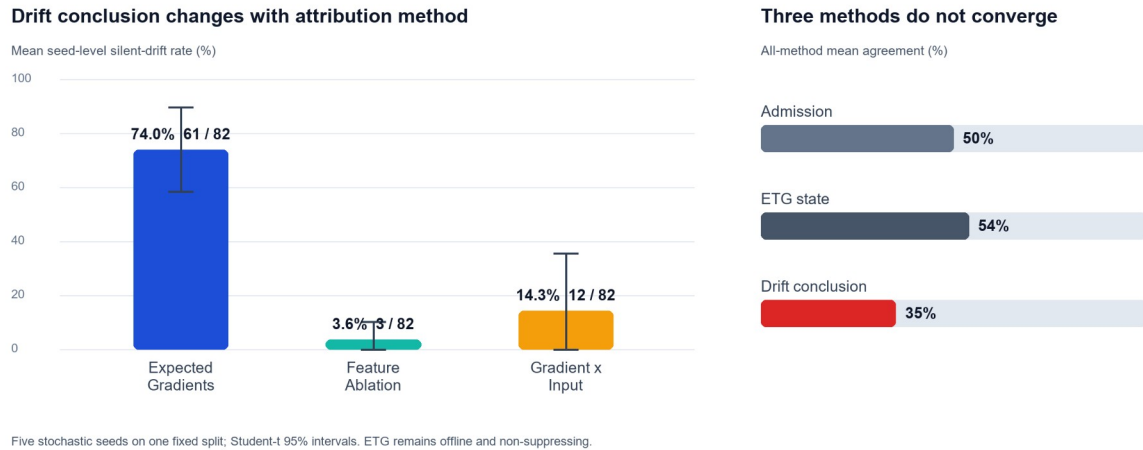


Figure 1. Five-seed attribution sensitivity and all-method agreement. Rates are method-conditioned governance evidence, not production incidents.

For Expected Gradients, the pooled five-seed ledger records 15 certified admissions, 35 refused admissions, 13 escalations, two strict recertifications, and six strict-recertification failures. Feature Ablation records 17 certifications and 33 refusals with no escalations, while Gradient x Input records six certifications, 44 refusals, one escalation, and one strict recertification. These differing state counts demonstrate execution of the same ledger under different explanations, but they also show that the governance outcome inherits the attribution choice. They do not quantify analyst benefit, false-alarm reduction, production remediation, or causal improvement of later OFRA checkpoints.

4.6 Exploratory open-set and labelled-head pilot

The separate seed-42 O1 pilot used FTP-Patator as an unseen class before its declared labelled increment. Thresholds were fitted on 11,647 training-only known-class calibration rows; the test set was not used for threshold selection. Confidence-only, centroid-distance-only, conservative joint, and empirical-joint rejection all produced 0% unknown recall. The best AUROC was 0.5303, the OSCR-style AUC was 0.4699, and the primary candidate buffer contained 509 known rows and no FTP-Patator rows. The registered gate therefore failed as an unknown-discovery mechanism.

After FTP-Patator labels were supplied at the legitimate task boundary, the ordinary supervised OFRA update achieved 82.30% new-class recall. Old-class accuracy changed from 87.28% to 87.01% (-0.27 percentage points). The evaluator never creates a semantic head automatically. This single-seed result supports labelled adaptation, not autonomous zero-day discovery, and remains an expected-contract reconstruction rather than an unchanged ReplayIDS reproduction.

Table 5. Seed-42 O1 open-set and post-label adaptation diagnostic.

Stage or metric	Result
Held-out class / test rows	FTP-Patator / 1,588
Pre-update known accuracy	87.28% (226,135 rows)
Unknown recall, all four gates	0.00%
Best AUROC / OSCR-style AUC	0.5303 / 0.4699
Post-label new-class recall	82.30%
Old-class accuracy change	-0.27 percentage points

4.7 Paired checkpoint-selection result

The training-only rule changes the operating trade-off but does not establish a stable improvement. Relative to last-epoch selection, final accuracy increases from 85.51% to 87.52% and Macro-F1 from 54.40% to 56.34%, while average task accuracy decreases from 79.78% to 77.88%, forgetting increases from 3.52% to 5.18%, and attack recall decreases from 85.73% to 83.76%. Benign FPR decreases from 13.10% to 10.44%. Every paired 95% confidence interval includes zero; the Holm-adjusted p-value for the two registered primary outcomes, Macro-F1 and forgetting, is 0.755. The candidate is therefore retained as an inconclusive ablation, and last-epoch selection remains the primary configuration.

Table 6. Per-class official joint-cap-3,000 results. Class-wise accuracy is recall; values are five-seed mean +/- sample SD on one fixed test split.

Class	Support	Last recall	Cal. recall	Last F1	Cal. F1
Benign	174,421	86.9% +/- 8.3%	89.6% +/- 5.0%	90.7% +/- 4.4%	92.1% +/- 2.8%
DoS GoldenEye	2,059	97.8% +/- 0.9%	94.7% +/- 4.8%	90.2% +/- 7.3%	92.5% +/- 2.0%
DoS Hulk	46,215	81.0% +/- 13.7%	81.1% +/- 18.1%	80.6% +/- 8.2%	82.7% +/- 7.5%
DoS Slowhttptest	1,100	75.0% +/- 14.3%	79.5% +/- 5.8%	54.6% +/- 7.7%	53.8% +/- 8.0%
DoS slowloris	1,159	69.0% +/- 21.2%	74.2% +/- 11.0%	48.3% +/- 16.5%	49.6% +/- 10.5%
FTP-Patator	1,588	69.0% +/- 44.3%	70.1% +/- 24.9%	30.9% +/- 28.5%	51.5% +/- 19.9%
Heartbleed	2	60.0% +/- 54.8%	60.0% +/- 54.8%	9.4% +/- 13.3%	7.6% +/- 10.4%
SSH-Patator	1,179	82.3% +/- 23.3%	69.0% +/- 38.7%	30.4% +/- 14.7%	21.0% +/- 16.6%

The per-class record explains why aggregate gains are not sufficient. Heartbleed has only two official test rows, so its mean and standard deviation are not a stable population estimate. FTP-Patator F1 improves under calibration, whereas SSH-Patator recall and F1 decline. The class-specific trade-offs reinforce the decision not to promote the candidate.

4.8 Guarded checkpoint-selection result

DICC Job 426307 repeated the five paired seeds with the recall/FPR guard. Relative to last epoch, guarded selection changed final accuracy from 85.51% to 87.10% (+1.59 points), Macro-F1 from 54.40% to 55.93% (+1.53), forgetting from 3.52% to 3.56% (+0.04), attack recall from 85.73% to 83.44% (-2.29), and benign FPR from 13.10% to 10.90% (-2.20). Average task accuracy decreased by 1.79 points. The Macro-F1 paired t-test gave $p=0.372$ and Holm-adjusted $p=0.744$; the forgetting test gave $p=0.977$. All paired 95% confidence intervals included zero. The guard removed the large mean forgetting penalty of the unconstrained selector but did not establish superiority, and only seed 2 Pareto-dominated its matched baseline across final accuracy, Macro-F1, forgetting, attack recall, and benign FPR. Last epoch therefore remains the formal primary rule.

Table 7. Guarded checkpoint selection versus the matched last-epoch baseline (five paired seeds).

Metric	Last epoch mean (SD)	Guarded mean (SD)	Paired change	95% CI
Average task accuracy	79.78 (6.86)	77.99 (10.01)	-1.79 pp	[-6.27, 2.69]
Forgetting	3.52 (2.60)	3.56 (4.71)	+0.04 pp	[-3.28, 3.36]
Final accuracy	85.51 (6.18)	87.10 (5.19)	+1.59 pp	[-0.33, 3.52]
Macro-F1	54.40 (5.07)	55.93 (5.02)	+1.53 pp	[-2.69, 5.75]
Attack recall	85.73 (14.98)	83.44 (17.49)	-2.29 pp	[-5.91, 1.33]
Benign FPR	13.10 (8.25)	10.90 (6.43)	-2.20 pp	[-4.87, 0.47]

4.9 RQ1 Same-data Balanced Replay50 comparison

Table 8 reports mean final metrics over five training seeds. Malaya OFRA has higher overall accuracy and lower signed forgetting in all five pairs. However, excluding screening seed 42, Macro-F1 is 20.70% versus 20.47% for replay, only +0.23

points; mean final task accuracy favours replay in all four new seeds. The large retention difference must therefore be read alongside weak absolute class-balanced performance.

Table 8. Same-data five-seed means. ReplayIDS uses last-epoch primary OFRA. Accuracy and recall values are percentages; forgetting is in percentage points.

Dataset / metric	Replay50	OFRA primary	Difference	Direction
Malaya accuracy	40.14	54.68	+14.54	Higher better
Malaya Macro-F1	18.72	21.15	+2.43	Higher better
Malaya balanced accuracy	22.13	23.03	+0.90	Higher better
Malaya signed forgetting	43.80	3.55	-40.25	Lower better
ReplayIDS accuracy	70.65	85.51	+14.86	Higher better
ReplayIDS Macro-F1	36.88	54.40	+17.52	Higher better
ReplayIDS balanced accuracy	87.22	77.63	-9.59	Higher better
ReplayIDS signed forgetting	5.89	3.52	-2.37	Lower better
ReplayIDS attack recall	99.78	85.73	-14.05	Higher better
ReplayIDS benign FPR	35.61	13.10	-22.51	Lower better

On ReplayIDS, primary OFRA improves overall accuracy and Macro-F1 and lowers benign FPR in all five pairs, but attack recall is lower in all five; forgetting improves in three. The test set is 76.59% benign, so fewer false alarms can increase overall accuracy while missing more attacks. Mean final task accuracy is 79.78% for OFRA versus 86.44% for replay. The four-seed sensitivity view retains the same central trade-off: accuracy 84.61% versus 71.71%, Macro-F1 54.07% versus 36.48%, attack recall 83.11% versus 99.77%, and FPR 13.76% versus 34.35%.

The same trained last-epoch OFRA states provide a mechanism diagnostic. Head-only Macro-F1 is 26.79%, router-only-cap-3,000 is 42.04%, and joint is 54.40%. Router-only attack recall is 92.06% versus joint 85.73%, while FPR falls from 21.76% to 13.10% when scores are combined. Thus routing contributes useful evidence, but score fusion changes the missed-attack/false-alarm balance. These retrospective scoring comparisons do not identify a unique causal component or justify optimising fusion weights on the test set.

4.10 ReplayIDS cross-environment score fidelity

A local CPU reconstruction checked all 20 stored primary checkpoint instances across five seeds against hash-bound historical GPU score snapshots. It used the original 512-row encoder chunks followed by whole-probe head/router scoring. The fixed gate required every score to satisfy absolute tolerance 0.0001 plus relative tolerance 0.00001 and required identical argmax predictions. Head scores passed at all checkpoints. The five initial checkpoints passed the combined gate, whereas all 15 post-initial Router/Joint checkpoints failed it. Maximum joint absolute error reached 0.01051009.

Two argmax differences occurred for the same probe row in seed 3 at checkpoints 001 and 003, changing DoS Hulk to Benign. They are two checkpoint evaluations of one row, not a full-test error rate or SHAP/ETG drift estimate. Reconstruction used CPU PyTorch 2.11.0 and NumPy 2.4.6; historical inference used A100 CUDA, PyTorch 2.6.0 and NumPy 2.2.6. The environments were not matched. Historical-environment GPU verification remains outstanding; no calibration was fitted, no historical score was replaced, and no revised full-test performance is claimed. This ReplayIDS result does not by itself re-evaluate the separate Malaya attribution campaign.

5. Discussion

5.1 When is the trade-off operationally worthwhile?

The value of routing depends on the dataset and operational cost model. NSL-KDD is favourable across the four displayed metrics. UNSW-NB15 trades lower accuracy and higher forgetting for better class-balanced metrics. CIC-IDS-2017 gains predictive coverage without a retention gain. Malaya improves coverage of minority application classes while its accuracy and forgetting differences remain uncertain. Such a trade may be worthwhile when a missed underrepresented class is more costly than a small aggregate-accuracy loss, but the decision must be based on per-class costs and cannot be inferred from overall accuracy alone.

The ReplayIDS replay comparison adds a distinct operational caution: OFRA reduces benign FPR but also misses more attacks in every paired seed. Whether that trade is acceptable depends on specified costs and minimum detection

requirements. No such deployment threshold or analyst benefit has been validated here, so higher accuracy is not sufficient grounds to recommend deployment.

5.2 Classifier capacity and class imbalance

The encoder and cumulative comparisons answer different questions. TabM-OFRA tests a different encoder within the existing head/router interface. Cumulative CatBoost and TabM estimate headroom when all seen-class rows are retained and retrained. TabM-OFRA improves aggregate accuracy but not class-balanced metrics, while cumulative CatBoost improves final classification metrics and shows greater checkpoint forgetting. These observations do not identify model capacity as the sole cause. The same-data replay comparison further shows that isolation, loss, sampling, and memory accounting must be separated before attributing gains to an architectural innovation.

5.3 Meaning of the integrated evidence chain

The current contribution is an integrated offline audit chain: OFRA generates the registered joint-cap decision; expected gradients explain its class margin; ETG applies declared evidence rules and records the outcome. Calling this a closed-loop OFRA-ETG system would be premature because ETG does not alter later model behaviour. The distinction matters for both scientific validity and system safety: post-hoc governance evidence can support review without being presented as a validated adaptive control mechanism.

5.4 Attribution dependence

The completed five-seed campaign confirms that ETG states inherit material attribution-method dependence. All-method agreement is 50.0% for admission, 54.0% for ETG state, and 35.0% for the drift conclusion. Expected Gradients remains the registered explainer because it is the predeclared differentiable method, not because the comparison proves it uniquely correct. A future operational design should require method identity in every ledger row and evaluate consensus, uncertainty-aware admission, or external calibration before any automated enforcement. The failed Integrated-Gradients completeness diagnostic remains a warning that a differentiable attribution method cannot be assumed valid for the complete piecewise routed score.

6. Limitations and Threats to Validity

First, strict five-seed prediction evidence is complete for all five datasets, but model capacity differs between Malaya and the four FT256x4 datasets. Second, the source-bound Malaya attribution and ETG analysis now covers five stochastic seeds on one fixed split; it does not establish cross-split or cross-device generalisation. Third, the three valid attribution methods disagree materially, and Integrated Gradients failed its earlier completeness diagnostic. Fourth, silent-drift thresholds are study-defined operating rules and have not been externally calibrated. Fifth, ETG actions are simulated and no analyst study or deployed feedback loop has been conducted.

Sixth, the current formal preprocessing uses Task-0-only numerical statistics and categorical vocabularies; the legacy 122/194-feature schema exposure was closed by strict cache rebuilding and affected five-seed reruns. Seventh, Malaya is not an intrusion-detection dataset, and its application labels cannot support attack-recall or benign false-positive claims. Eighth, Malaya is strongly imbalanced and several minority recalls remain low even after routing. Ninth, cumulative CatBoost and TabM diagnostics are not matched continual-learning baselines. Tenth, complete runtime, explanation latency, energy, analyst burden, and protocol-matched external-baseline tests are not yet available across the full dataset suite. Eleventh, the FTP-Patator open-set result is a single-seed, single-held-out-class diagnostic; its 0% unknown recall invalidates the current gate but does not establish that all open-set methods will fail. Twelfth, the paired checkpoint-selection experiment uses five seeds on one fixed split; it does not establish cross-split generalization, and the Heartbleed calibration fallback prevents a fully uniform class rule.

Two further boundaries affect the newer comparison. ReplayIDS D2's training cap uses future-class fitting counts for offline construction, even though fitted normalisation is Task-0-only. The CPU reconstruction also fails the registered Router/Joint numerical-fidelity gate in a different runtime; source and checkpoint hashes alone therefore do not establish portable numerical equivalence. Same-environment verification is needed before further score calibration or attribution work on that ReplayIDS path. The balanced-replay results compare complete training systems with unmatched total memory and exposure and cannot isolate a human-designed component's causal contribution.

7. Conclusion

OFRA exhibits dataset-dependent retention and detection trade-offs across five independently processed datasets. The same-data replay comparison strengthens the Malaya retention finding but shows little Macro-F1 improvement in the four non-screening seeds. On ReplayIDS, accuracy and Macro-F1 gains coexist with lower attack recall, and numerical reconstruction in a different runtime does not pass the fixed fidelity gate. The Malaya attribution study finds strongly method-dependent drift and governance states. ETG can record this uncertainty without suppressing prediction, but is not a validated safety certificate or online controller. The contribution is an explicitly testable prediction-and-audit design and its empirical limits; architecture-only superiority, portable score identity, and deployment benefit remain unproven.

Declarations

Funding: No external funding was declared for this study.

Conflict of interest: The author declares no conflict of interest.

Ethics and data governance: The study uses previously collected public benchmark or research datasets and does not involve direct interaction with human participants. Dataset licences and institutional computing rules remain applicable. No restricted credentials or private institutional data are included in the public repository.

Code and data availability: Reproducibility code, configurations, preprocessing contracts, result summaries, and public evidence bindings are available at <https://github.com/1838904818/Class-IL>. Original datasets remain with their respective providers; the repository documents acquisition and preparation rather than redistributing data where licence or size makes redistribution inappropriate.

AI assistance: AI-based tools assisted code development, technical analysis, and manuscript drafting and revision. These tools are not authors. Responsibility for the research decisions, verification of results, interpretation, and final manuscript remains with the author. The contribution claims concern the specified design and empirical evidence, not an assertion that all constituent methods or ideas were created without assistance.

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